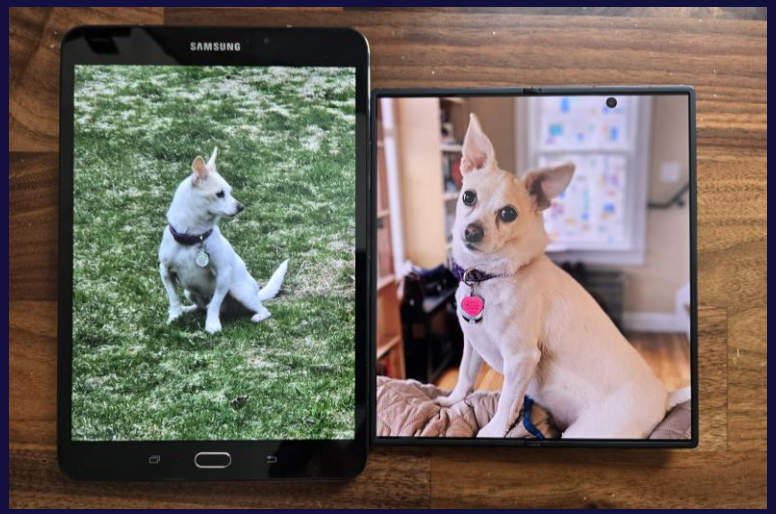


Change Tracking in SQL Server 2025: Exploring Change Event Streaming vs CDC

Deborah Melkin
(she\her)
Data Engineer

About Me

- 25+ years as a DBA
- Data Platform Women in Tech (WIT) Virtual UG, Co-leader
- WITspiration, Co-founder
- Redgate Community Ambassador
- Microsoft MVP, Data Platform

- In my spare time, I can usually be found doing something musical or something geeky with my husband, Andy, and our dog, Sebastian.

Here's the Scenario:

- Breaking up a legacy monolith system into smaller parts
  - Microservices
  - Many databases
  - Teams choosing their database RDBMS & set ups
- Creation of analytical system
  - (Near) Real-time analytics
- No cross-database joins
- Need to share data across the systems
- Need before and after data changes

Options We're Ruling Out

Option 1: Add columns to the table

```sql
CREATE TABLE dbo.Users(
    Id INT IDENTITY(1,1) NOT NULL,
    DisplayName NVARCHAR(40) NOT NULL,
    CreationDate DATETIME NOT NULL,
    LastUpdateDate DATETIME NOT NULL,
    CONSTRAINT PK_Users_Id PRIMARY KEY CLUSTERED (Id),
    INDEX IDX_Users_DisplayName (DisplayName)
    )
```

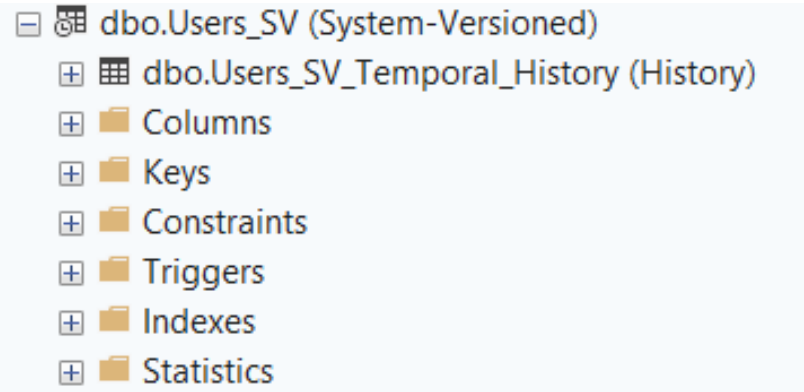
Pros:

- We can use these columns directly

Cons:

- Querying large tables can take time
- Not guaranteed to always be populated properly
- Only has the updated data

Option 2: Temporal Tables\System Versioned Tables



Pros:

- Maintained by SQL Server
- Stores both previous and current versions of rows

Cons:

- Querying large tables can take time
- Potential overhead if not set up properly with additional indexing and retention periods.
- See documentation for limitations

Option 3: Change Tracking

Pros:

- Only identifies that row that has changed
- Managed by SQL Server

Cons:

- Does not capture the original or updated data
- Requires additional set up and overhead.

What are we left with?

Change Data Capture (CDC)

- Introduced SQL Server 2008
- “Pull”

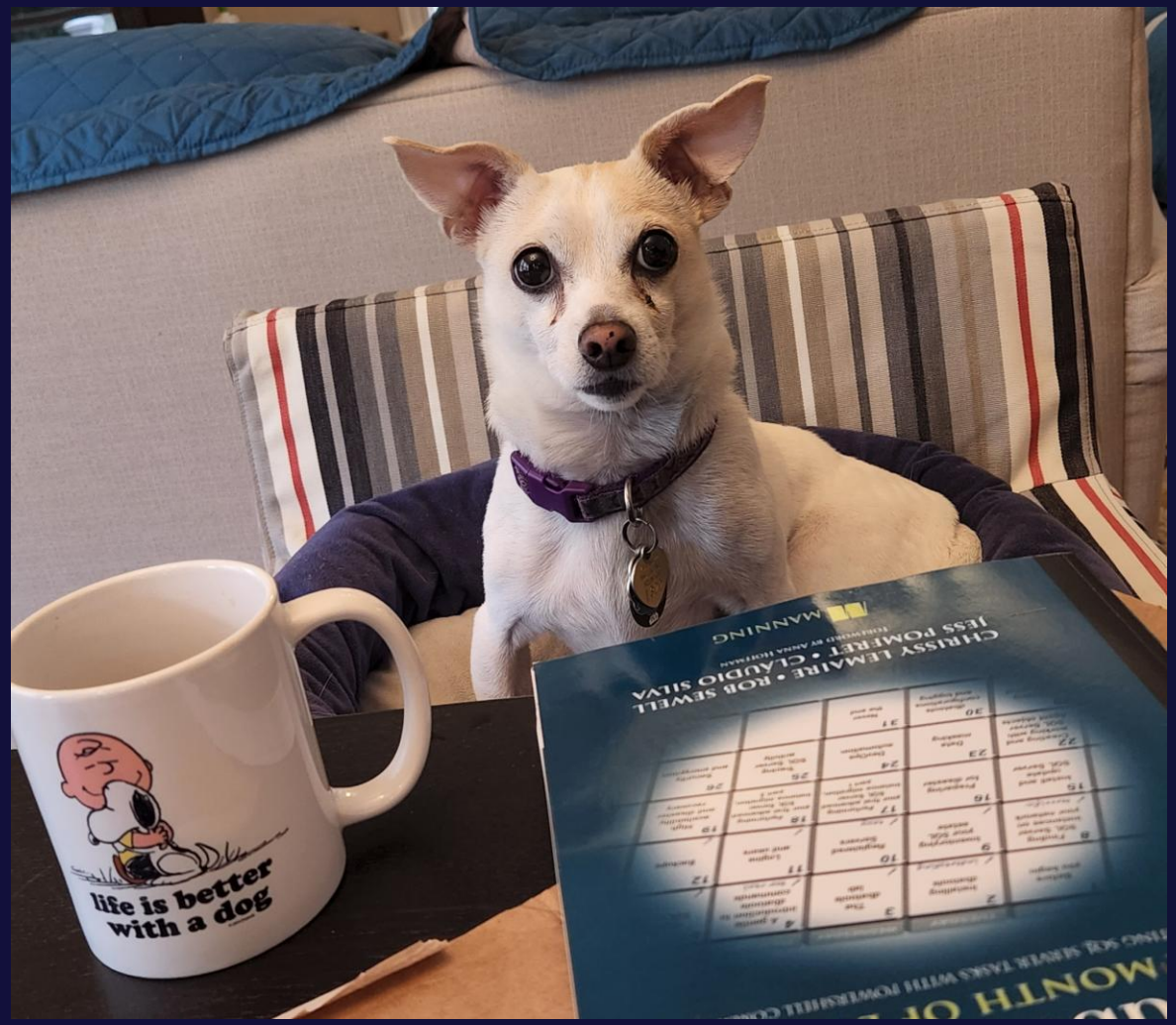
Change Event Streaming (CES)

- Introduced SQL Server 2025
- Still in Preview
- “Push”

Agenda

- How CDC & CES Work
- Setup
- Demos
- “Cautionary Tales”
- When To Use Each

How They Work



Overview of Capture Process for CDC & CES

- Specify tables to track
- Transaction Logs are scanned for changes which are sent to the appropriate system
  - CDC – cdc schema tables inside the database
  - CES – Azure Event Hubs
    - Managed, high-throughput message service
- Logs receive acknowledgement of changes
- After retention period, change data is removed

How this works:

```
UPDATE Customer
SET State = 'MA'
WHERE FirstName = 'Sebastian'
```

Fake\Mock Transaction Log

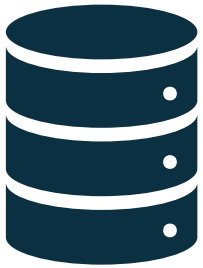
Log Seq Number (LSN)	Table	Record Info	Chg	Clear?
0000000A	Customer	Sebastian-AZ	Old	✓
0000000A	Customer	Sebastian-MA	New	✓

Acknowledgement

OR

Changes Sent

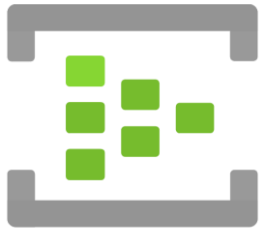
cdc schema tables



CDC

CES

Azure Event Hubs



Transaction Logs – Word of Warning

- Can't clear logs if changes can't be written or messages haven't been acknowledged.

- Monitor Log Growth to make sure that these are not blockers:

```
SELECT db.name,
    db.is_cdc_enabled,
    db.is_change_feed_enabled,
    db.log_reuse_wait_desc,
    usage.total_log_size_in_bytes/1024 AS TotalSizeInKB,
    usage.used_log_space_in_bytes/1024 AS TotalSpaceInKB,
    usage.used_log_space_in_percent,
    usage.log_space_in_bytes_since_last_backup/1024 AS SpaceUsedSinceLastBackupInKB
FROM sys.databases AS db
JOIN sys.dm_db_log_space_usage AS usage ON usage.database_id = db.database_id
```

REPLICATION

Setup

CDC

Azure Event Hubs

CES

CDC Set up

- Enable CDC in the database
- Enable CDC for the tables
- SQL Server Agent must be running
  - 2 jobs are created per database: Capture & Cleanup
  - Scheduler for Azure SQL DB

Agent Job Activity:

Name	Enabled	Status	Last Run Outcome	Last Run
cdc.StackOverflow_CDC_capture	yes	Executing...	Unknown	never
cdc.StackOverflow_CDC_cleanup	yes	Idle	Succeeded	3/4/2026 3:07:43 AM

CDC SQL Agent Jobs – Capture

- Settings for job procs
  - MaxTrans = # of transactions in each scan cycle (Default 10000)
  - Maxscans = # of scan cycles to execute to get all the rows (Default 10)
  - Continuous = Is this running continuously? (Default 1)
  - PollingInterval = # of seconds between scans (Default 5)
- Must be running for CDC to be working
- Starts when SQL Agent Job starts by default
  - Consider adding an additional job schedule to start if it is stopped
  - Add check to monitoring\alerting system to see if it’s not running

CDC SQL Agent Jobs – Cleanup

- Settings for job procs
  - Retention Period = How long to keep records (Default 4320 minutes\3 days)
  - Threshold = Batch size (Default 5000)

- Deletes in batches:

```
delete top( @p1 )  from [cdc].[dbo_Users_CT]
where __$start_lsn < @p2
```

  - Performance Consequence if not run in a while

- SQL Server – defaults to once a day
- Azure SQL DB - Runs every hour while database is running

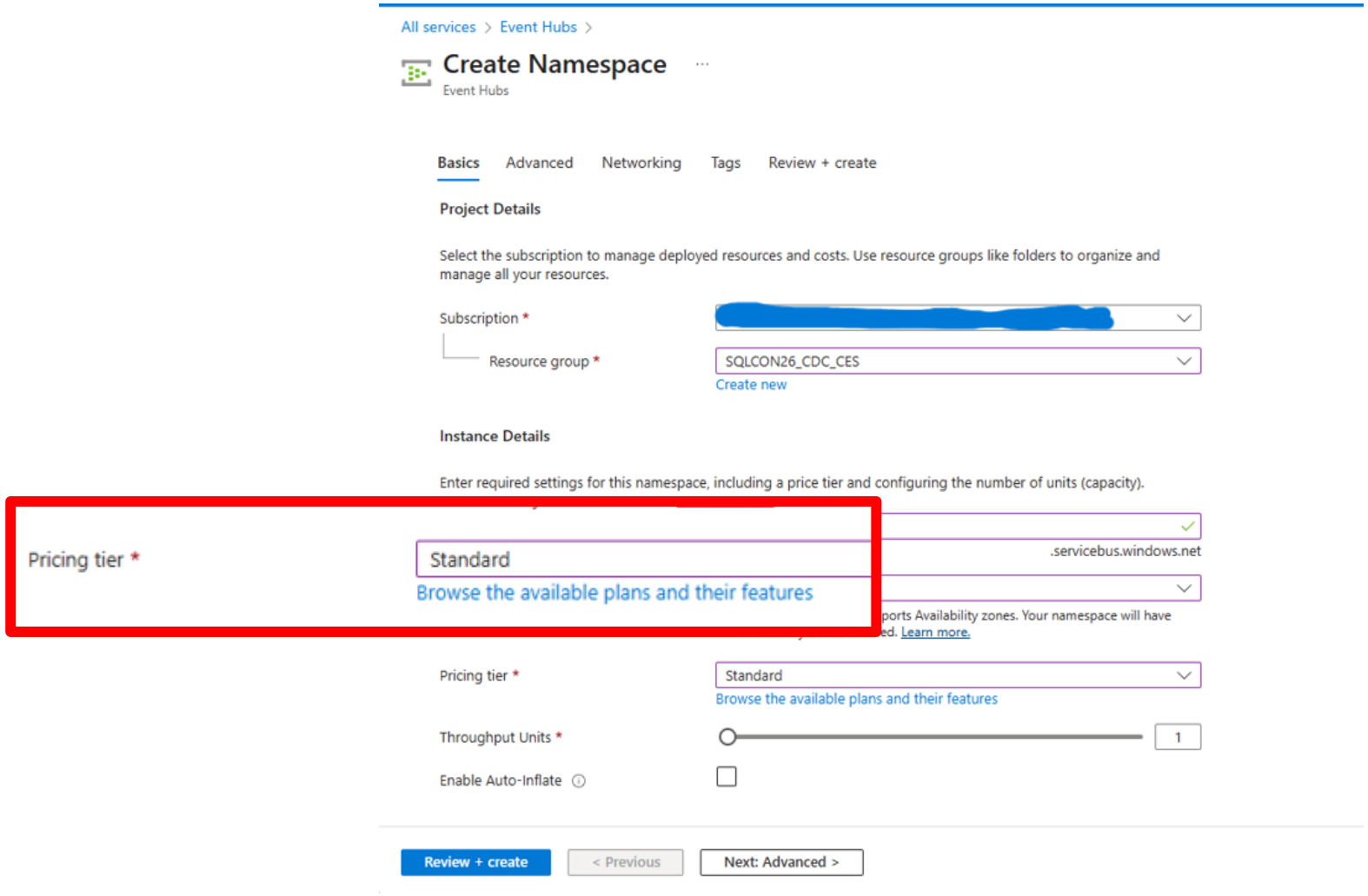
CES Set up - Azure Event Hubs

- Must be set up before CES is enabled
- Choose protocol for the messages
  - AMQP
  - Kafka
- Security currently differs depending on your SQL engine
  - EntraID (Azure SQL DB)
  - Security Access Key (SQL Server)
- Can have multiple partitions for messages in Event Hub

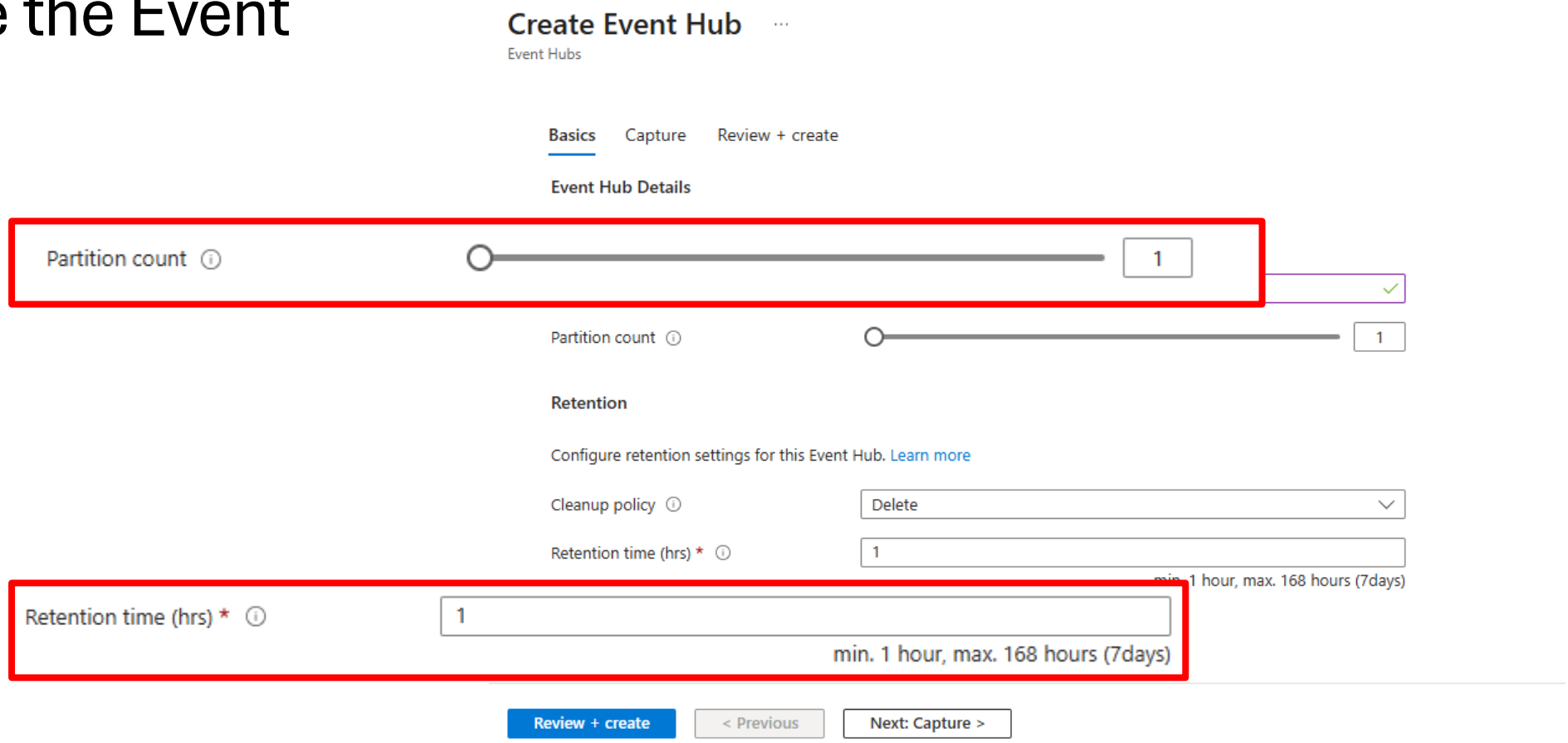
1. Create the Namespace

Pricing Tiers:

- Basic
- Standard
- Premium
- Dedicated



2. Create the Event Hub



3. Create the Shared Access Policy – AMQP Protocol

Skip if using EntraID

(Azure SQL DB)

https://learn.microsoft.com/en-us/sql/relational-databases/track-changes/change-event-streaming/configure?view=sql-server-ver17&tabs=sas-access%2Csas-token-auth

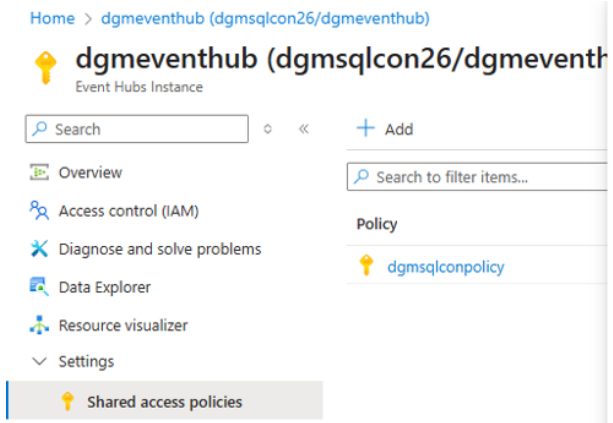
```
PS C:\> function Generate-SasToken {
>> $subscriptionId = "<Azure-Subscription-ID>"
>> $resourceGroupName = "<Resource-group-name>"
>> $namespaceName = "<Azure-Event-Hub-Namespace-name>"
>> $eventHubName = "<Azure-Event-Hubs-instance-name>"
>> $policyName = "<Policy-name>"
>> $tokenExpireInDays = "<number-of-days-token-will-be-valid>"
>>
>> # Modifying the rest of the script is not necessary.
>>
>> # Login to Azure and set Azure Subscription.
>> Connect-AzAccount
>>
>> # Get current context and check subscription
>> $currentContext = Get-AzContext
>> if ($currentContext.Subscription.Id -ne $subscriptionId) {
>>     Write-Host "Current subscription is $($currentContext.Subscription.Id), switching to $subscr
iptionId..."
>>     Set-AzContext -SubscriptionId $subscriptionId | Out-Null
>> } else {
>>     Write-Host "Already using subscription $subscriptionId."
>> }
>>
>> # Try to get the authorization policy (it should have Send rights)
>> $rights = @("Send")
>> $policy = Get-AzEventHubAuthorizationRule -ResourceGroupName $resourceGroupName -NamespaceName $
namespaceName -EventHubName $eventHubName -AuthorizationRuleName $policyName -ErrorAction SilentlyC
ontinue
>>
>> # If the policy does not exist, create it
>> if (-not $policy) {
>>     Write-Output "Policy '$policyName' does not exist. Creating it now..."
>>
>> # Create a new policy with the Manage, Send and Listen rights
>>     $policy = New-AzEventHubAuthorizationRule -ResourceGroupName $resourceGroupName -NamespaceNa
me $namespaceName -EventHubName $eventHubName -AuthorizationRuleName $policyName -Rights $rights
```

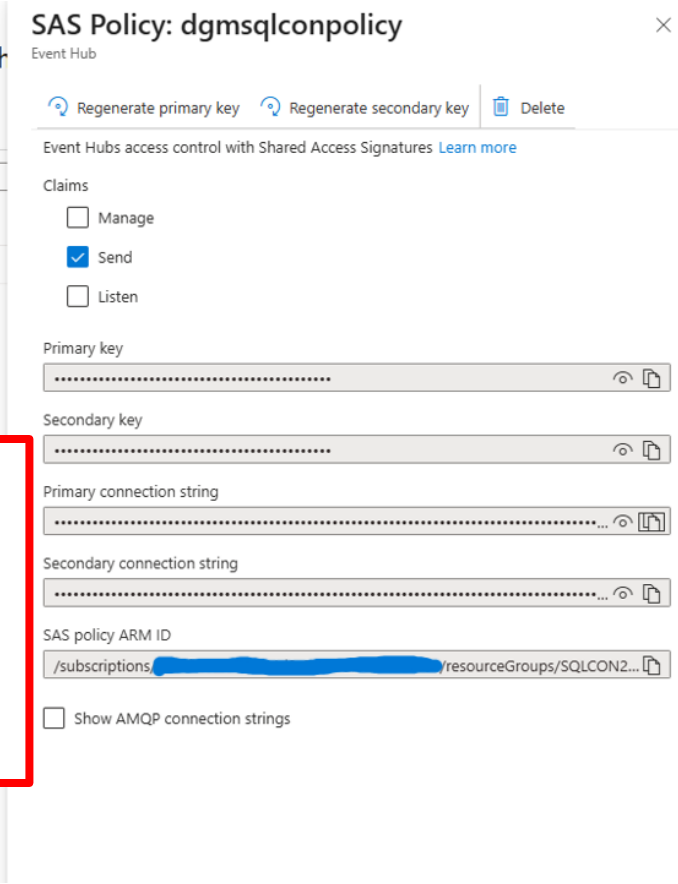
```
-- Generated SAS Token --
SharedAccessSignature sr=https%3a%2f%YourEventHubNamespace.servicebus.windows.net%2fYourE
xxxx&se=2059133074&skn=SharedKeyNameIsHERE
-- End of generated SAS Token --
```

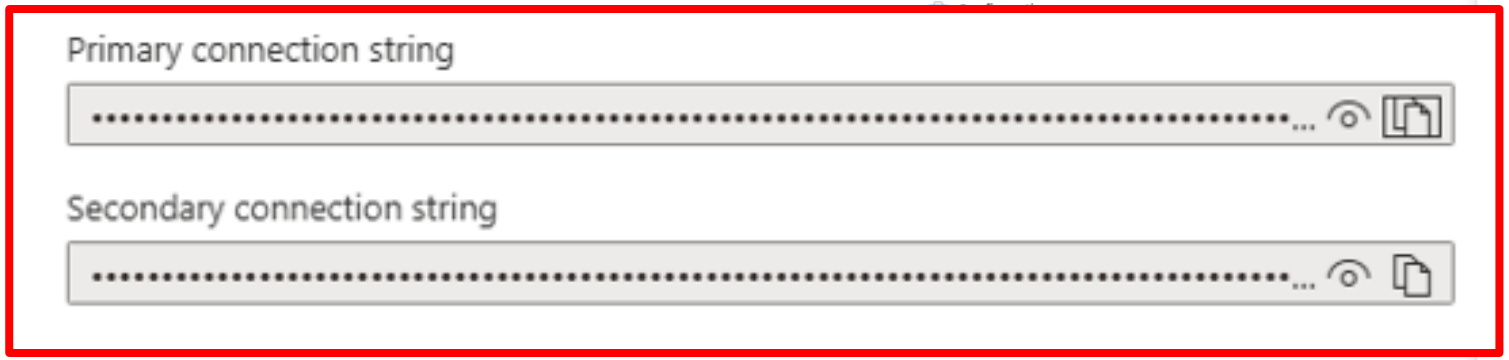
3. Create the Shared Access Policy – Kafka Protocol

Skip if using EntraID

(Azure SQL DB)







CES Set up - Database

- Enable Preview Features as this is technically still in Preview
  - Not needed for Azure SQL DB
- Enable Event Stream for the database
- Create an Event Stream Group for sending the messages
- Add the table to the stream group
  - Tables can only be in one stream group

NOTE: CES is incompatible with:

- CDC, DB Mirroring in Fabric, Replication

Demos

- Set up database
- Make changes

Cautionary Tales

aka Problems to Look For

High Availability – Availability Groups

- Neither are AG aware
- Consider making modifications to the CDC jobs:
  - Add a job step to make sure that the database is the primary replica.
    - If it is, continue as normal. Otherwise, end job.
  - Add the jobs to the replicas in advance so they are ready to kick off if/when a failover occurs.
- If you are using read replicas to get LSNs for processing, the LSNs could be out of sync if the AGs are out of sync.

Disaster Recovery

- Restores
  - CDC requires an additional command for cdc to be retained: WITH KEEP_CDC
  - CES restores do not keep CES settings
- Disabling CDC erases everything!

Processing Messages (CES)

- Ordering of events isn't necessarily maintained
  - May need additional help to identify the order of the updates
  - Within a partition, the logicalid value may be able to help
- Duplicate Messages are possible
  - If an acknowledgment isn't received for some reason, SQL Server will keep retrying the send
  - Consumer should be aware and know how to handle these cases
- Learning Curve
  - Azure, JSON, .NET, Kafka, Stream Analytics Jobs, etc.

Data Initialization

- CDC and CES are about capturing and streaming changes
- May require full resets at some point for some reason
  - New Setup
  - “Breaking” Schema Changes
  - Disaster Recovery
- Have a plan for reseeding the data for the target systems

Money! Money! Money!

- CES & Azure Event Hubs
  - Pricing Guide: Azure Event Hubs Pricing
  - Make sure you have the right tier set up
  - Know what happens when you hit limits
  - Emulator (unsupported) for development purposes:
    - MS Learn - Azure Event Hubs Emulator Overview

- CDC on Azure SQL DB
  - I/O of writing to and deleting from cdc schema tables
  - Transaction logs

When to use each

When to use CDC

- Changes do not need to be changed outside the environment
  - Same database
  - Different database, same SQL Server instance
  - Different SQL Server instance, same network
- Batch processing
- No Azure or other cloud service
- Need Replication, DB Mirroring in Fabric as well
- SQL Server 2022 or earlier (Can still use with SQL 2025)

When to use CES

- Distributed systems
  - Azure DBs only
  - Hybrid Cloud & On Prem
  - Different RDBMS
- Real-time/near real-time processing
  - “Streaming data”

Resources

Transaction Log Resources

- MS Learn - SQL Server Transaction Log Architecture and Management Guide
- Paul Randal Blog Series:
  - https://www.sqlskills.com/blogs/paul/category/transaction-log/
  - https://www.sqlskills.com/blogs/paul/the-sql-server-transaction-log-part-1-logging-basics/

CDC Resources

- MS Learn - About Change Data Capture SQL Server
- MS Learn - Change Data Capture Overview (Azure SQL DB)
- MS Learn - Enable and Disable Change Data Capture - SQL Server
- MS Learn - Administer and Monitor Change Data Capture
- MS Learn - System Tables for Change Data Capture
- MS Learn - System Stored Procs - Change Data Capture

CES & Azure Event Hubs Resources

- MS Learn - Change Event Streaming Overview
- MS Learn - Change Event Streaming – Configure
- MS Learn - Azure Event Hubs - Create
- MS Learn - Azure Event Hubs Emulator Overview
- Azure Event Hubs Pricing

Any Questions?

Thank You For Coming!

Github: github.com/DebtheDBA/CDC_vs_CES

Blog: DebtheDBA.wordpress.com

BlueSky: @dgmelkin.bsky.social

How was the session?

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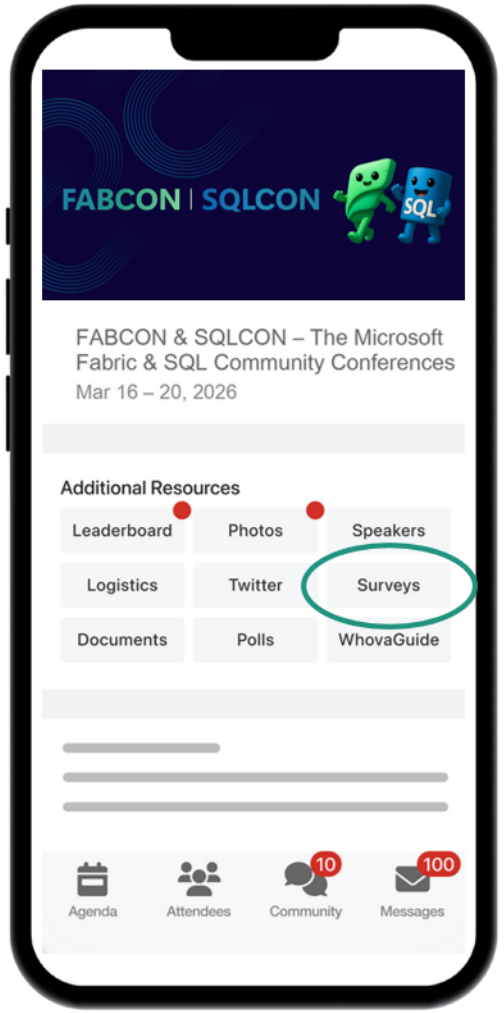

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